

FDG Fidelity Audit

This is the polished audit for the FDG Typst conversion. It replaces the sequential working notes as the main planning document while preserving those notes at:

- `audit-snapshots/2026-07-03-rough-audit-baseline/audit-rough.typ`
- `audit-snapshots/2026-07-03-rough-audit-baseline/audit-rough.pdf`

The main purpose is to guide fixes efficiently. It focuses on known issues, where to repair them, what target each region should follow, and what evidence must be preserved as the PDFs reflow.

Source provenance note: `fdg-book/` is treated as a hard-copied vendored source tree, not a moving subtree dependency. Its import details and update policy are recorded in `fdg-book/PROVENANCE.md`.

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Executive Summary

Area	Status	Decision
Layout	ACCEPTED	US-letter layout is intentional. Page-for-page reproduction of the published PDF is not a goal. Audit content loss, bad breaks, obscured text, and incorrect counters instead.
References	PARTIAL	Reference numbering is seeded to match the published PDF. Remaining work is formatting and every callout/entry check.
Footnotes	BUG	Published footnote numbering is a fidelity requirement. Current global Typst numbering drifts from published chapter/appendix numbering.
Equations	BUG	Several source/PDF math drifts are known. Equation counter drift also breaks downstream references.
Code/results	PARTIAL	Many code blocks are present; executable verification remains incomplete, especially errata claims and high-density chapters.
Figures	PARTIAL	Imported PDF crops are acceptable for now but should probably be redrawn in CeTZ over time.
Index	BUG	Generated index exists but differs from the published index in symbol entries, locator semantics, and explanatory note.
Errata	PROJECT-ADDED	Intentional project end matter. Needs wording/code cleanup and executable verification for uncertain claims.

Canonical Targets

Use the right target before filing or fixing a mismatch.

Region	Canonical target	Notes
Vendored source	<code>fdg-book/</code> hard copy	<code>fdg-book/PROVENANCE.md</code> records the subtree import commits and freezes this directory as ordinary tracked repo content. Do not

		refresh from upstream during fidelity work without a new snapshot and reviewed diff.
Book body prose/math/code	Published PDF first, Org second	When Org and PDF differ, classify as source drift. Repair in converter or source according to the fix policy below.
Bibliography numbering	Published PDF	Numbering/order should match published references. Formatting is still under review.
Footnote numbering	Published PDF	Published numbering is now a fidelity requirement. This affects chapter and appendix references.
Layout geometry	Project policy	Letter paper is intentional; only audit readability/content effects.
Chapter 11 reconstructed equations	Published PDF	Org has empty equation environments in several places. These are intentional PDF-derived repairs.
Errata	Org errata plus executable checks	Errata is not in the published PDF. Review it as project end matter.
Index	Hybrid	Compare vocabulary/grouping/styling against published index. Exact locators are not required while pagination differs.
Figures	Published visual target, Typst-native future	Current assets are PDF crops. Future direction is CeTZ redraws with controlled text layers.

Evidence Preservation

The first audit snapshot has been created:

```
audit-snapshots/2026-07-03-rough-audit-baseline/
```

It contains the current `fdg-book.pdf`, the rough audit PDF/source, the published PDF copy, representative raster evidence, and `MANIFEST.md`. Any rough-audit citation of current **physical page N** should be interpreted as referring to that snapshot unless a later row explicitly re-verifies the page after fixes.

Before each fix pass:

1. Snapshot the current rendered PDFs and any raster evidence pages.
2. Record `git rev-parse HEAD`, dirty status, build commands, and PDF metadata.
3. Fix the issue.
4. Rebuild `fdg-book.pdf` and `typ/audit.pdf`.
5. Add post-fix verification without deleting the old evidence trail.

Evidence commands used for the baseline snapshot:

```
typst compile audit-snapshots/2026-07-03-rough-audit-baseline/audit-rough.typ audit-
snapshots/2026-07-03-rough-audit-baseline/audit-rough.pdf
node scripts/assert-typst-regressions.mjs
pdffinfo fdg-book.pdf
pdffinfo audit-snapshots/2026-07-03-rough-audit-baseline/audit-rough.pdf
pdftoppm -f 163 -singlefile -png -r 130 fdg-book.pdf audit-snapshots/2026-07-03-rough-audit-baseline/
rasters/current-index-pl63
pdftoppm -f 166 -singlefile -png -r 130 fdg-book.pdf audit-snapshots/2026-07-03-rough-audit-baseline/
rasters/current-errata-pl66
pdftoppm -f 240 -singlefile -png -r 130 fdg-book/fdg_book.pdf audit-snapshots/2026-07-03-rough-audit-
baseline/rasters/published-index-p240
pdftoppm -f 152 -singlefile -png -r 130 fdg-book.pdf audit-snapshots/2026-07-03-rough-audit-baseline/
rasters/current-appendix-b-pl52
pdftoppm -f 220 -singlefile -png -r 130 fdg-book/fdg_book.pdf audit-snapshots/2026-07-03-rough-audit-
baseline/rasters/published-appendix-b-p220
```

Fix Policy

Fix site	Use when
----------	----------

Org source	The Org is clearly wrong, the correction belongs upstream, and regeneration should preserve it. Example: source math drift such as Appendix C equation C.1.
Converter	A class of Org patterns needs systematic conversion or generated Typst needs stable repairs after regeneration. Example: manual equation alignment markers, double-parenthesized equation references, and PDF-derived reconstruction rules.
Typst helper/lib	The content is correct but presentation/counters/layout are wrong across many generated files. Example: footnote numbering and figure rendering.
Generated Typst	Only for temporary diagnosis. Avoid direct generated-file fixes unless they are immediately moved into Org/converter/helper code.
Audit only	The issue is accepted styling, project policy, or a warning that should not alter the book.

Every fix should add or update a regression when the behavior can be checked cheaply.

Priority Repair Queue

Pri	Status	Issue	Repair direction and evidence
P0	BUG	Footnote numbering	Published footnote numbering is required. Current Typst footnotes continue globally; examples include Appendix A first note rendering as 108 instead of 1. Repair likely belongs in <code>typ/fdg-lib</code> footnote/chapter handling. Recheck chapter 8 because footnote display equations currently perturb equation numbering and references.
P0	BUG	Equation counter drift	Chapter 8 display equations inside footnotes cause later equation numbers/references to drift; Appendix C currently says torsion is equation (8.23) while published says 8.21. Fix shared counter behavior before local cross-reference patches. Evidence: rough entries C5-012 and C7-012.
P1	SOURCE-DRIFT	Appendix B equation B.7	Published page 220 has $I_0(s)=t$; Org <code>fdg-book/scheme/org/appendix_b.org:244-253</code> and generated <code>typ/content/appendix_b.typ:143-151</code> render $I_0(s)=y$. Repair from PDF.
P1	SOURCE-DRIFT	Appendix B equation B.10	Published page 222 labels the product AC; Org <code>appendix_b.org:338-343</code> and generated <code>appendix_b.typ:199-201</code> label the left side AB. Repair from PDF.
P1	SOURCE-DRIFT	Appendix C equation C.1	Published page 232 has second term $g T(v, \omega)$; Org <code>appendix_c.org:19-22</code> and generated <code>appendix_c.typ:8-14</code> repeat u . Repair in Org or converter with explicit PDF-derived note.
P1	SOURCE-DRIFT	Appendix C equation C.10	Published page 234 has primed Riemann component on the left; Org <code>appendix_c.org:129-133</code> and generated <code>appendix_c.typ:79-81</code> put the unprimed component there. Repair with nearby C.8-C.10 notation in mind.
P1	BUG	Chapter 10 equations 10.3 and 10.6	Equation 10.3 has a sign/formula mismatch against published page 175. Equation 10.6 needs both formula repair and manual converter-side alignment characters to match the PDF. Evidence: rough entries C6-006 to C6-009.
P1	BUG	Chapter 11 double equation references	Current page 131 renders $((11.11))$; generated <code>typ/content/chapter011.typ:68</code> inherits already-parenthesized source. Fix reference rewriting in <code>scripts/convert-org-to-typst.mjs:1160-1196</code> rather than patching one file.
P2	BUG	Index note and symbol entries	Published index page 240 says procedure definition locators are italic and n marks footnotes. Current page 163 says generated entries include procedure definitions and lacks the opening symbol-entry block. Fix <code>typ/index.typ:13-390</code> or document limitations explicitly.
P2	BUG	Copyright/CIP front matter	Known visual details still need repair: missing CC BY-NC-SA banner; <code>special_sales@mitpress.mit.edu</code> raw/monospace; extra vertical space

			before Library of Congress line; indent before p. cm.; wider print-line spacing; em dash before dc23; right-align 2012042107. Source: <code>typ/fdg-lib/title.typ:40-69</code> .
P2	PARTIAL	References formatting	Numbering/order now match samples via seeded bibliography order. Exhaustively compare all callouts and all 21 bibliography entries; decide whether IEEE-style formatting is acceptable.
P3	PARTIAL	Cover art tuning	Fine-tune the custom cover in <code>typ/fdg-lib/cover.typ</code> : the particle trajectory should run against the edge of the potential wells rather than floating too far inside/away from them. A 3D diagram/scene could be considered if it communicates the geometry cleanly, but this is low priority compared with content/counter fixes.
P3	PARTIAL	Errata wording/code	Keep errata as project-added, but adjust low-confidence wording and fix suggested code names. <code>literal-oneform-field</code> should be <code>literal-lform-field</code> ; suggested literal symbols should be quoted.
P3	PARTIAL	Figures	Current PDF crops are visually serviceable but create text-layer extraction artifacts. Plan CeTZ redraws figure by figure after content-critical fixes.

Complete Actionable Inventory

This table reconciles the rough audit against this polished audit. Every rough row marked **BUG**, **SOURCE-DRIFT**, or **PDF-REPAIR** should appear here, even when it is lower priority or already covered by a broader repair theme.

Rough ID	Status	Area	Action
C1-010	BUG	Front matter	Add missing CC BY-NC-SA banner/mark near the license paragraph in <code>typ/fdg-lib/title.typ:43</code> .
C1-011	BUG	Front matter	Render only <code>special_sales@mitpress.mit.edu</code> as raw/monospace; keep <code>creativecommons.org</code> in main text font.
C1-013	BUG	Front matter	Increase vertical space before Library of Congress Cataloging-in-Publication Data.
C1-014	BUG	Front matter	Indent the p. cm. line in CIP data.
C1-015	BUG	Front matter	Increase spacing in the final 10 9 8 7 6 5 4 3 2 1 print line.
C1-016	BUG	Front matter	Use an em dash before dc23.
C1-017	BUG	Front matter	Right-align 2012042107.
Cover art	PARTIAL	Front matter	Fine-tune the custom cover art in <code>typ/fdg-lib/cover.typ</code> : the particle trajectory should run against the edge of the wells. Optional 3D diagram exploration is allowed but low priority.
C2-003	BUG	References	Current IEEE bibliography style does not match published prose; prototype manual visible bibliography or custom renderer.
C2-004	BUG	References	Restore published author phrasing, including with Julie Sussman and with Meinhard E. Mayer.
C2-005	BUG	References	Entry [6] must render the Dover/New York/1989 note.
C2-006	BUG	References	Entry [10] must render IEEE organization and standard number.
C2-007	BUG	References	Entry [18] must use published memo wording: Artificial Intelligence Laboratory memo AIM-2002-018, November 2002.
C2-008	BUG	References	Entry [21] must use published prose Free software is available at: followed by URL.
C2-009	BUG	References	Book entries should follow published publisher/address/year order rather than current IEEE ordering.
C2 entry matrix	BUG	References	All 21 entries remain individually actionable for prose/style comparison: [1]-[21] in rough lines C2 matrix.

C3-005	BUG	Prologue	Preserve Papert citation [13] inside prologue footnote; likely converter citation handling inside footnotes.
C3-006	SOURCE-DRIFT	Prologue	Compare and possibly repair lowercase <code>this lagrangian</code> source drift against published prose.
C3-008	BUG	Chapter 1	Normalize $90^\circ/90^\circ$ E degree glyph and spacing in longitude prose.
C3-010	BUG	Chapter 1	Fix chapter 1 footnote numbering as part of global published-footnote policy.
C4-003	BUG	Chapter 3	Remove literal hyphenation artifacts such as <code>di- rection</code> , <code>deriva- tives</code> , and <code>com- ponents</code> via converter cleanup.
C4-004	BUG	Chapter 3	Fix chapter 3 footnote numbering as part of global published-footnote policy.
C4-005	BUG	Chapter 3	Fix equation-reference drift and double-parentheses near one-form discussion; published target is (3.41), not ((3.43)).
C4-007	BUG	Chapter 4	Repair equation 4.6 layout so $c_j^k(m)$ stays with the main displayed equation before the number.
C4-008	BUG	Chapter 4	Recheck/fix references to equations 3.40/3.41 after chapter 3 equation label repair.
C4-014	BUG	Chapter 5	Prevent footnote display equations from advancing visible body equation numbers.
C4-015	SOURCE-DRIFT	Chapter 5	Repair equation 5.24 commutator term $\omega([v_i, v_j], \dots)$ and add manual converter-side alignment chars.
C5-004	SOURCE-DRIFT	Chapter 6	Repair <code>make fake-vector-field</code> to <code>make-fake-vector-field</code> .
C5-005	SOURCE-DRIFT	Chapter 6	Repair <code>coordinate-system at</code> to <code>coordinate-system-at</code> .
C5-006	SOURCE-DRIFT	Chapter 6	Repair pullback parameter mismatch <code>f-on-m</code> vs <code>f-on-M</code> .
C5-007	SOURCE-DRIFT	Chapter 7	Repair prose typo <code>asume</code> to <code>assume</code> .
C5-012	BUG	Chapter 8	Fix footnote display-equation counter drift that shifts body equations 8.10-8.12.
C5-013	SOURCE-DRIFT	Chapter 8	Repair <code>However ,</code> to <code>However,.</code>
C5-014	SOURCE-DRIFT	Chapter 8	Repair symmetric-connection condition to show lower indices swapped on the right: $\Gamma^{i_{jk}} = \Gamma^{i_{kj}}$.
C6-006	PDF-REPAIR	Chapter 10	Preserve converter repairs for malformed Hodge-dual source text around g^{ij} and epsilon subscript.
C6-007	SOURCE-DRIFT	Chapter 10	Repair equation 10.3 so $\text{grad}(f) = g^{\text{sharp}}(df)$ has no trailing (df) .
C6-008	BUG	Chapter 10	Improve equations 10.5-10.6 display layout with manual alignment/line-break repair.
C6-009	SOURCE-DRIFT	Chapter 10	Repair equation 10.6 second curl component basis form from dx to published dy .
C6-011	PDF-REPAIR	Chapter 11	Preserve PDF-derived reconstructions for equations 11.1-11.14 and 11.20-11.21.
C6-012	PDF-REPAIR	Chapter 11	Preserve intentional <code>Einsten</code> to <code>Einstein</code> repair.
C6-013	PDF-REPAIR	Chapter 11	Preserve Lorentz interval repair from repeated ξ^2 to ξ^1 .
C6-014	BUG	Chapter 11	Fix double-parenthesized equation reference $((11.11))$.
C6-015	PDF-REPAIR	Chapter 11	Preserve code repairs for <code>make-4tuple</code> , $R^{(-1)}$, and norm notation.

C7-004	BUG	Appendix A	Fix appendix footnote numbering as part of global published-footnote policy.
C7-006	SOURCE-DRIFT	Appendix B	Repair equation B.7 selector result to $I_0(s)=t$.
C7-007	SOURCE-DRIFT	Appendix B	Repair equation B.10 left side to AC.
C7-010	SOURCE-DRIFT	Appendix C	Repair equation C.1 second term to $g T(v, \omega)$.
C7-011	SOURCE-DRIFT	Appendix C	Repair equation C.10 primed/unprimed left side.
C7-012	BUG	Appendix C	Recheck torsion reference after chapter 8 equation-counter fix; do not patch locally first.
C8-005	BUG	Index	Fix index explanatory note and procedure-definition locator policy.
C8-006	BUG	Index	Add/restore published opening symbol and notation index entries.
C8-008	BUG	Index	Reduce/generated index granularity or document policy; current entries are too mechanical versus curated published index.

Unchecked Inventory

These are the explicit rough-audit **UNCHECKED** items and completion markers. They are not necessarily defects; they are the remaining audit surface.

Rough ID	Area	Unchecked work
C2-010	References	Visually verify title styling/italics in the bibliography; <code>pdftotext</code> cannot decide this.
C2-F01	References	Prototype/evaluate custom CSL or bibliography-style route.
C2-F05	References	Prototype recommended manual visible <code>fdg-published-bibliography()</code> plus hidden bibliography for citation resolution.
C3-014	Chapter 2 figures	Raster-compare figures 2-1, 2-2, and 2-3 for placement, scale, caption punctuation, and adjacency.
C3-015	Preface/prologue/chapters 1-2	Complete normalized section-level text diffs for every paragraph, footnote, equation, Scheme block, result block, exercise, figure caption, and cross-reference.
C4-017	Chapters 3-5	Check every equation number/reference target, all footnote numbers/text, all Scheme blocks/results, all exercises, all figure visual crops/captions, and normalized paragraph text.
C5-016	Chapters 6-8	Check every equation number/reference target, all published footnote numbers/text, every Scheme block/result, all exercises, all citations, all section-title line breaks, and figure scaling.
C6-016	Chapters 9-11	Check every equation symbol and alignment, all code/result blocks, all footnote numbers/text, all citations, every cross-reference, and all exercises.
C7-014	Appendices A-C	Check every Scheme block/result pair, all footnote text and numbering, every equation symbol/alignment, all appendix citations, and all appendix cross-references back into the main text.
C8-014	Index/errata	Check every published index term/subterm, alphabetical/symbol sort order, every code/procedure locator, footnote <i>n</i> locators, procedure-definition italic locators, every errata sentence, and every errata code block.

Region Findings

Front Matter

Status: **PARTIAL** with accepted layout differences.

Accepted:

- US-letter layout and custom cover/title treatment are intentional.

- Errata as end matter is intentional.
- `typ/main.pdf` and `fdg-book.pdf` should differ only by cover/index material.

Actionable details:

- Add the CC BY-NC-SA banner.
- Render `special_sales@mitpress.mit.edu` in raw/monospace.
- Keep `creativecommons.org` in main text font.
- Keep stylistic LaTeX baseline offsets.
- Add vertical space before Library of Congress Cataloging-in-Publication Data.
- Add indent before `p. cm.`.
- Increase spacing in the `10 9 8 7 6 5 4 3 2 1` print line.
- Use an em dash before `dc23`.
- Right-align `2012042107`.
- Fine-tune the custom cover art in `typ/fdg-lib/cover.typ`: the particle trajectory should run against the edge of the potential wells. Consider a 3D diagram/scene only if it improves the geometric reading without becoming a distraction; this is low priority.
- Add a final visual QA pass for cover/title/end-matter variants after content-critical fixes, because subjective polish issues such as cover trajectory placement are easy to miss in text extraction.

Evidence:

```
pdftotext -f 1 -l 10 fdg-book/fdg_book.pdf -
pdftotext -f 1 -l 10 fdg-book.pdf -
nl -ba typ/fdg-lib/title.typ | sed -n '35,75p'
nl -ba typ/fdg-lib/cover.typ | sed -n '1,120p'
```

References And Citations

Status: **PARTIAL**.

The reference-number problem has been mostly addressed by seeding bibliography order. Published references begin with SICP as [1], and current samples now match the important callouts. Remaining work is a complete callout/entry audit and a style decision for generated IEEE formatting.

Evidence:

```
pdftotext -f 230 -l 239 fdg-book/fdg_book.pdf -
pdftotext -f 160 -l 162 fdg-book.pdf -
nl -ba typ/fdg-lib/refs.typ | sed -n '1,180p'
nl -ba typ/content/references.typ | sed -n '1,80p'
```

Regression wishlist:

- Assert SICP is [1].
- Assert Turtle Geometry is [2].
- Assert Scheme standard is [10].
- Assert Spivak is [17].
- Add one text assertion per chapter with citations once numbering is stable.

Footnotes And Counters

Status: **BUG**.

Published footnote numbering is a fidelity requirement. Current numbering is global across the book. This creates visible mismatches in appendices and likely in chapters. More importantly, chapter 8 footnote display equations affect equation counters and corrupt downstream references.

Repair direction:

- Inspect `typ/fdg-lib/chapter.typ`, `typ/fdg-lib/refs.typ`, and any footnote show rules.
- Decide whether footnotes reset per chapter/appendix or follow the exact published pattern.
- Ensure display equations inside footnotes do not consume main equation numbers unless the published PDF does.

- After fixing, recheck Appendix C torsion reference to equation 8.21.

Evidence:

```
pdftotext -f 143 -l 145 fdg-book.pdf -
pdftotext -f 206 -l 208 fdg-book/fdg_book.pdf -
pdftotext -f 159 -l 161 fdg-book.pdf -
pdftotext -f 232 -l 236 fdg-book/fdg_book.pdf -
rg -n 'footnote|counter|equation' typ/fdg-lib typ/lib.typ scripts/convert-org-to-typst.mjs
```

Equations And Source Drift

Status: **BUG** and **SOURCE-DRIFT**.

Highest-confidence PDF/source math mismatches:

Item	Source anchors	Repair note
B.7	appendix_b.org:244-253 appendix_b.typ:143-151	253 published page 220: $I_0(s)=t$, not y .
B.10	appendix_b.org:338-343 appendix_b.typ:199-201	343 published page 222: left side AC, not AB.
C.1	appendix_c.org:19-22 appendix_c.typ:8-14	22 published page 232: second term should contain v , not repeated u .
C.10	appendix_c.org:129-133 appendix_c.typ:79-81	133 published page 234: primed component belongs on the left side.
10.3	chapter010.org:56-60 chapter010.typ:25-29	60 visual PDF check found sign/formula mismatch. Reopen with published page 175 before patching.
10.6	chapter010.org:101-104 chapter010.typ:46-62	104 second curl component should be on dy ; also requires manual converter-side alignment chars.

Evidence:

```
pdftotext -f 152 -l 153 fdg-book.pdf -
pdftotext -f 220 -l 222 fdg-book/fdg_book.pdf -
pdftotext -f 159 -l 161 fdg-book.pdf -
pdftotext -f 232 -l 236 fdg-book/fdg_book.pdf -
pdftotext -f 117 -l 119 fdg-book.pdf -
pdftotext -f 174 -l 176 fdg-book/fdg_book.pdf -
```

Code Blocks And Executable Claims

Status: **PARTIAL**.

The audit has many text/visual checks, but code/result fidelity needs executable verification. Errata items in particular should be categorized as one of:

- hard runtime bug
- chapter-isolation setup dependency
- version-sensitive scmutils behavior
- mathematical/source error
- optional explanatory note

Executable verification plan:

1. Pin the scmutils version or commit used for testing.
2. Record the Scheme preload/setup file for each chapter.
3. Run errata claims as independent transcripts.
4. Store transcripts in an audit evidence directory.
5. Promote high-confidence behaviors into regression tests where feasible.

Errata review highlights:

- Keep hard runtime issues such as S^2 vs S^2 -type, missing R2-rect-point, missing V, missing SR-basis, and undefined eta-inverse.
- Rewrite Cartan order as historical if current Org now defines the block before use.
- Rewrite literal-oneform-field to literal-lform-field.
- Quote suggested literal symbols: use 'omega and 'f.
- Verify the-ether against the installed scmutils library before claiming it as the preferred home definition.

Evidence:

```
nl -ba fdg-book/scheme/org/errata.org | sed -n '1,340p'  
rg -n 'literal-oneform-field|literal-lform-field|eta-inverse|S\\^2|S\\^2-type|the-ether|home' fdg-book  
typ
```

Figures

Status: **PARTIAL**.

Current Typst uses imported PDF assets:

```
typ/assets/figures/fig-2-1.pdf  
typ/assets/figures/fig-2-2.pdf  
typ/assets/figures/fig-2-3.pdf  
typ/assets/figures/fig-4-1.pdf  
typ/assets/figures/fig-4-2.pdf  
typ/assets/figures/fig-5-1.pdf  
typ/assets/figures/fig-6-1.pdf  
typ/assets/figures/fig-7-1.pdf
```

These are visually useful, but they cause text-layer artifacts in **pdftotext** because embedded figure text is extracted alongside generated captions. The likely long-term direction is CeTZ redraws so figure geometry, labels, text layers, and captions are native Typst source.

CeTZ migration plan:

Figure	Priority	Plan
2.1-2.3	MEDIUM	Audit first; redraw once chapter 2 text/equations are stable.
4.1-4.2	MEDIUM	Good candidates for CeTZ because they are geometric basis/dual-basis diagrams.
5.1	MEDIUM	Redraw after integration section fidelity is stable.
6.1	LOW	Keep crop unless visual mismatch appears.
7.1	MEDIUM	Likely useful CeTZ candidate because it supports covariant-derivative intuition.

Evidence:

```
rg -n 'fdg-figure|fig-' typ/content typ/fdg-lib/figures.typ  
pdftoppm -f 41 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p41  
pdftoppm -f 54 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p54  
pdftoppm -f 82 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-p82
```

Index

Status: **BUG**.

Comparison policy: exact page locators do not need to match while geometry differs. Compare vocabulary, grouping/subentries, ordering, symbol/code styling, footnote n locators, and procedure-definition locator styling.

Known gaps:

- Published index opens with symbol and notation entries before A. Current generated index starts with section 4, then A/B/C.

- Published note says procedure definition pages are italic and n marks footnotes. Current note says procedure definition pages are included with generated entries.
- Current generated entries are broader and more mechanical for common code terms such as `define`, `down`, `compose`, and `differential`.

Repair direction:

- Inspect `typ/index.typ:13-390`.
- Determine whether `in-dexter` can style definition locators or whether a manual/custom index layer is needed.
- Add explicit symbol entries or a manual prelude if generated matching cannot cover the published opening block.

Evidence:

```
nl -ba typ/index.typ | sed -n '1,430p'
pdftotext -f 163 -l 165 fdg-book.pdf -
pdftotext -f 240 -l 249 fdg-book/fdg_book.pdf -
pdftoppm -f 163 -singlefile -png -r 130 fdg-book.pdf /tmp/fdg-current-index-p163
pdftoppm -f 240 -singlefile -png -r 130 fdg-book/fdg_book.pdf /tmp/fdg-published-index-p240
```

Errata

Status: PROJECT-ADDED and PARTIAL.

Errata is intentional project end matter, included after the generated index. It should be compared against `fdg-book/scheme/org/errata.org`, generated `typ/content/errata.typ`, and executable `scmutils` behavior, not against the published PDF.

Items to change before treating errata as polished:

Item	Score	Action
Cartan order	3/5	Current Org defines the named block before <code>geodesic-equation-residuals</code> ; make this historical/published-order wording or remove for current source.
omega definition	3/5	Change <code>literal-oneform-field</code> to <code>literal-1form-field</code> and quote <code>'omega</code> .
f definition	5/5	Keep but quote <code>'f</code> .
S ² issues	5/5	Keep; likely hard runtime bugs.
J vs circular	5/5	Keep; local source defines <code>circular</code> , then later calls <code>J</code> . Also inspect nearby <code>d/x</code> .
Simplifier/performance notes	2-3/5	Keep only with <code>scmutils</code> version, machine, timeout, and transcript.
home/the-ether	3/5	Verify <code>the-ether</code> in installed <code>scmutils</code> before asserting.

Evidence:

```
nl -ba fdg-book/scheme/org/errata.org | sed -n '1,340p'
nl -ba typ/content/errata.typ | sed -n '1,280p'
pdftotext -f 166 -l 170 fdg-book.pdf -
```

Regression Wishlist

Area	Assertions to add
References	Reference order and selected callouts for [1], [2], [10], [17].
Footnotes	First footnote number per chapter/appendix; no main equation counter increments from footnote-only display equations unless published does so.
Equation drift	Text assertions for B.7, B.10, C.1, C.10, 10.3, and 10.6 after fixes.
References/crossrefs	No double-parenthesized equation references such as <code>((11.11))</code> .
Figures	All 8 figure assets or future CeTZ figures are present; generated captions do not duplicate visibly.
Cover	Cover trajectory touches/reads against the edge of the wells; any optional 3D version is visually checked across desktop/PDF export and retained only if clearer than the 2D cover.

Index	Index note matches chosen policy; representative symbol entries exist; representative code entries exist.
Errata	Errata generated source uses <code>literal-lform-field</code> , quoted literal symbols, and expected chapter headings.

Work Plan

Recommended order:

1. Preserve snapshot before every fix pass.
2. Fix counter infrastructure first: footnotes and equation numbering.
3. Fix high-confidence source/PDF math drift: B.7, B.10, C.1, C.10, 10.3, 10.6.
4. Fix converter-level reference rewriting, including double-parenthesized equation refs.
5. Re-audit references formatting and all citation callouts.
6. Polish front matter details.
7. Fine-tune cover art and run a final visual QA pass for cover/title/end-matter variants.
8. Clean errata wording and run executable errata verification.
9. Rework index policy/implementation.
10. Start CeTZ figure migration after content-critical issues are stable.

Commit Trail

The rough audit recorded the historical trail through:

```
693bae8 2026-07-03 cleanup
9512cbe 2026-07-03 draft as default + makefile
413cbff 2026-07-02 fixes
a1006ad 2026-07-02 fmt
c9e554f 2026-07-02 add license, typos
cfa1f68 2026-07-02 Merge commit 'e1481748...' as 'fdg-book'
e148174 2026-07-02 Squashed 'fdg-book/' content from commit 9beb699
9edbac7 2026-07-02 Initial commit
```

This polished audit was created at HEAD:

```
6cb615bc4e14759678b0e7720efbbedbc8748a92
```